

Robert Riley Nerem

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EDUCATION

Doctor of Philosophy, Data Science (in progress) September 2022 → Present

University of California San Diego, San Diego, CA

Advisor: Prof. Yusu Wang, Halıcıoğlu Data Science Institute

Master of Science, Physics

May 2022

University of Calgary, Calgary, AB

Supervisor: Prof. Barry Sanders, Department of Physics and Astronomy

Co-supervisor: Prof. Peter Høyer, Department of Computer Science

Bachelor of Science, Physics

December 2018

Montana State University, Bozeman, MT

Honors Degree, *summa cum laude*

Relevant minors: Computer Science, Mathematics

PUBLICATIONS

Conference Papers

S. Wittig, A. Vasileiou, *R.R. Nerem*, T. Stoll, F. Geerts, Y. Wang, and C. Morris. [Which Algorithms Can Graph Neural Networks Learn?](#), International Conference on Machine Learning (ICML), 2026.

R.R. Nerem, S. Chen, S. Dasgupta, and Y. Wang. [Graph Neural Networks Extrapolate Out-of-Distribution for Shortest Paths](#), Proceedings of the Thirty-Ninth Conference on Learning Theory, PMLR **336**: 5273-5331, 2026.

R.R. Nerem, Z. Luo, A. Rafiey, and Y. Wang. [Differentiable Extensions with Rounding Guarantees for Combinatorial Optimization over Permutations](#), Advances in Neural Information Processing Systems 38 (NeurIPS 2025), Main Conference Track, 2025.

A.B. Kahng, *R.R. Nerem*, Y. Wang, and C.-Y. Yang. [NN-Steiner: A Mixed Neural-Algorithmic Approach for the Rectilinear Steiner Minimum Tree Problem](#), Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence, Article 1452, 24 March 2024.

D.V. Reddy, *R.R. Nerem*, A.E. Lita, S.W. Nam, R.P. Mirin, and V.B. Verma. [Exceeding 95% System Efficiency Within the Telecom C-band in Superconducting Nanowire Single Photon Detectors](#), Proceedings of CLEO: QELS Fundamental Science, San Jose, United States of America, 5 May 2019 - 10 May 2019.

Journal Articles

R.R. Nerem and D. Gaur. [Conditions for Advantageous Quantum Bitcoin Mining](#), Blockchain: Research and Applications, **4**(3): 100141, 1 September 2023.

A. Alase, *R.R. Nerem*, M. Bagherimehrab, P. Høyer, and B.C. Sanders. [Tight Bound for Estimating Expectation Values from a System of Linear Equations](#), Phys. Rev. Research, **4**: 023237, 2022.

P. Crawford-Kahrl*, *R.R. Nerem**, B. Cummins, and T. Gedeon. [Genetic Networks Encode Secrets of Their Past](#), Journal of Theoretical Biology, **541**(111092), 21 May 2022. * indicates co-first authorship.

D.V. Reddy, *R.R. Nerem*, S.W. Nam, R.P. Mirin, and V.B. Verma. [Superconducting Nanowire Single-photon Detectors with 98% System Detection Efficiency at 1550 nm](#), Optica, **7**(12): 1649-1653, 23 November 2020.

L.M. Smith, F. Motta, and 12 others including *R.R. Nerem*. [An Intrinsic Oscillator Drives the Blood Stage Cycle of the Malaria Parasite, Plasmodium falciparum](#), Science, **368**(6492): 754-759, 15 May 2020.

R.R. Nerem and D. James. [Charged Particle Pickup Tube Detector used in Experimental Applications and Classroom Demonstrations](#), The Physics Teacher, **58**(3): 200-205, 19 February 2020.

E. Berry, B. Cummins, *R.R. Nerem*, L. Smith, S. Haase, and T. Gedeon. [Using Extremal Events to Characterize Noisy Time Series](#), Journal of Mathematical Biology, **80**: 1523-1557, 1 February 2020.

Preprint

R.R. Nerem, P. Crawford-Kahrl, B. Cummins, and T. Gedeon. [A Poset Metric from the Directed Maximum Common Edge Subgraph](#), arXiv:1910.14638, October 2020.

RESEARCH EXPERIENCE

Computation

PhD Research

Fall 2022 → Present

Prof. Yusu Wang

Halcioğlu Data Science Institute

University of California San Diego, San Diego, CA

- Developed neural networks aligned with classical algorithms for chip-design routing and related combinatorial optimization problems
- Derived theoretical bounds explaining when aligned GNNs extrapolate on shortest path problems
- Created differentiable extensions for combinatorial optimization over permutations with provable rounding guarantees

Quantum Researcher

Fall 2021 → Fall 2022

Quantum Algorithms Institute

Surrey, BC

- Analyzed quantum machine learning algorithms for medical biology

Master's Research

Fall 2019 → May 2022

Prof. Barry Sanders & Prof. Peter Høyer

Institute for Quantum Science and Technology

University of Calgary, Calgary, AB

- Developed quantum algorithms for estimating expectation values, yielding end-to-end quantum algorithms for solving systems of linear equations

- Proved lower bounds for expectation value problems by reducing from oracle problems with known lower bounds
- Worked closely with a group in biology to determine applicability of machine learning quantum algorithms to pathogen identification problems in medical biology
- Evaluated implications of quantum search algorithms to the security of Bitcoin proof-of-work

Research Assistant

Fall 2018 → Summer 2019

Prof. Tomas Gedeon & Dr. Breschine Cummins, Mathematics

Montana State University, Bozeman, MT

- Developed a poset metric based on maximum common subgraphs to characterize the distance between gene expression time series
- Constructed algorithms to find the maximum common edge subgraph between two node labeled digraphs
- Determined graph structures that are conserved under evolution by vertex duplication, and developed algorithms to identify such structures
- Created Python framework to simulate genetic network evolution and compare resultant graphs with random graph models

Experiment

Research Assistant

Summer 2018

Dr. Sae Woo Nam, Faint Photonics Group

National Institute of Standards and Technology, Boulder, CO

- Precisely measured efficiency of superconducting nanowire single-photon detectors (SNSPD) using extensive use of Python programming to automate measurements
- Automated measurement of integrated photonic devices using Python to directly interface with layout design software

Research Assistant

Fall 2016 → Spring 2018

Prof. Rufus Cone & Dr. Charles Thiel, Physics

Montana State University, Bozeman, MT

- Developed a series of low-cost scanning Fabry-Perot interferometers to evaluate laser performance
- Used absorption and fluorescence spectrum measurements along with spectral hole burning to determine suitability of europium and thulium doped crystals for quantum information applications

Research Assistant

Summer 2015, Summer 2016

Prof. Zoltan Sternovsky & Dr. David James

Institute for Modeling Plasma, Atmospheres and Cosmic Dust

University of Colorado, Boulder, CO

- Fully developed and characterized a pickup tube detector to measure charge and velocity of micron-sized particles in the accelerator housed at IMPACT
- Integrated an array of photomultiplier tubes, including design of high-speed PCB trans-impedance amplifiers, to record acceleration profiles of ablating high-velocity dust particles incident on a gas target